

Factor each completely. Show your work for partial credit.

1. $25x^2 - 49y^2$	2. $6x^3 - 18x^2 + 5x - 15$
Factored Form	Factored Form
3. $24x^5y^6z^4 + 54x^8y^4z^9$	4. $16x^2 + 72x + 81$
Factored Form	Factored Form
5. $x^2 - 5x - 14$	6. $9x^2 + 21x + 10$
Factored Form	Factored Form

7. $2x^2 + x - 36$		8. $x^2 + 7x - 30$	
Factored Form		Factored Form	
9. $x^6 - 25$		10. $x^2 + 20x + 99$	
Factored Form		Factored Form	
11. $5x^2 - 4x + 20x^3 - 1$		12. $24x^2y - 16xy + 56y^3$	
Factored Form		Factored Form	

13. $x^2 - 10x + 21$	14. $8x^3y^2 + 56x^7y^5 + 32x^6y^4$
Factored Form	Factored Form
15. $x^2 + 16x + 64$	16. $8x^2 + 22x - 21$
Factored Form	Factored Form

17. An expression is shown.

$$8x^3 - 12x^2 - 2x + 3$$

Select all of the equivalent expressions.

- ☐ $2x(4x^2 - 6x - 1)$
- ☐ $2x^2(4x - 6x - 1)$
- ☐ $4x^2(2x - 3) - 1(2x + 3)$
- ☐ $4x^2(2x - 3) - 1(2x - 3)$
- ☐ $(2x - 3)(2x - 1)(2x + 1)$